Programming Assignment: Implement k-Means from Scratch

Objective

Implement the k-means clustering algorithm **from scratch** in Python, without using any built-in k-means implementation (for example, `sklearn.cluster.KMeans`). You may use standard libraries such as `numpy` for array and mathematical operations and `matplotlib` for displaying images and plots.

Your implementation must include the following subroutines with appropriate inputs:

- `initialize_centroids`
- `assign_clusters`
- `update_centroids`

Part 1: Clustering random 2D points

1. Generate 1000 random points in the 2D plane, where each coordinate (x, y) is uniformly sampled in the range $[0, 500]$.
2. Run your k-means algorithm with $k = 4$ for the following initial centroid configurations:
   a. Initialize the centroids at $(125, 125)$, $(125, 375)$, $(375, 375)$, and $(375, 125)$. Display the final clusters.
   b. Initialize the centroids at $(50, 50)$, $(50, 125)$, $(50, 250)$, and $(50, 375)$. Display the final clusters.
   c. Initialize the centroids at four randomly chosen points from the dataset. Display the final clusters.

   For each configuration, record the number of iterations required for convergence.

Part 2: Clustering pixels in a handwritten image

3. Using any image-editing tool, create a black-and-white image of the first three letters of your name.
   - Each letter should have a width and height between 100 and 200 pixels.
   - The spacing between two adjacent letters should be 100 pixels.
4. Treat each pixel in the resulting image as a 2D data point (represented by its coordinates).
5. Run your k-means algorithm with $k = 3$ for the following initial centroid configurations:
   a. Initialize the centroids at $(125, 125)$, $(125, 375)$, and $(375, 375)$. Display the final clusters.
   b. Initialize the centroids at $(50, 50)$, $(50, 125)$, and $(50, 250)$. Display the final clusters.

c. Initialize the centroids at three randomly chosen points from the pixel coordinates. Display the final clusters.
d. Initialize the centroids at one point sampled from each of the three letters. Display the final clusters.

For each configuration, record the number of iterations required for convergence and comment briefly on any observed differences.